A New Auxiliary Model Approach For Fermionic Criticality

Insights on Mottness in Strongly Correlated Systems

ABHIRUP MUKHERJEE

Presubmission Open Seminar

IISER KOLKATA

Emergent Phenomena in Quantum Matter group
IISER Kolkata

EPQM

Acknowledgements

Publications

New J. Phys. 25 113011. **AM**, N S Vidhyadhiraja, A Taraphder, S Lal. (2023)

J. Phys. A: Math. Theor. 57 275401. **AM**, S Patra, S Lal. (2024)

New J. Phys. 28 013503. **AM**, S R. Hassan, A Mukherjee, N S. Vidhyadhiraja, A Taraphder, S Lal. (2026)

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in progress. **AM**, S Lal. (2026)

Phys. Rev. B 105, 085119. A Mukherjee, **AM**, N S. Vidhyadhiraja, A Taraphder, S Lal. (2022)

J. Phys.: Condens. Matter 35 315601. S Patra, **AM**, A Mukherjee, N S Vidhyadhiraja, A Taraphder, S Lal. (2023)

Phys. Rev. B 111, L140414. A Kumar N M, S Mukherjee, **AM**, A Punjal, S Purwar, T Setti, S Prabhu S., S Lal, N Kamaraju. (2025)

arXiv:2602.20600. D Debata, **AM**, S Lal. (2026)

Outline of Thesis

Chap 1. Introduction

Chap 2. Methods and Preliminaries

Chap 3. Kondo frustration via charge fluctuations: Mott localisation in $d = \infty$ [New J. Phys. 25 113011]

Chap 4. A New Auxiliary Model Mapping For Correlated Lattice Models [New J. Phys. 28 013503]

Chap 5. Mott Criticality as the Confinement Transition of a Pseudogap-Mott Metal in $d = 2$ [New J. Phys. 28 013503]

Chap 6. Competing tendencies In Heavy Fermion Systems: Kondo screening vs. Mott localisation [in progress]

Chap 7. Holographic entanglement renormalisation for fermionic quantum matter [J. Phys. A: Math. Theor. 57 275401]

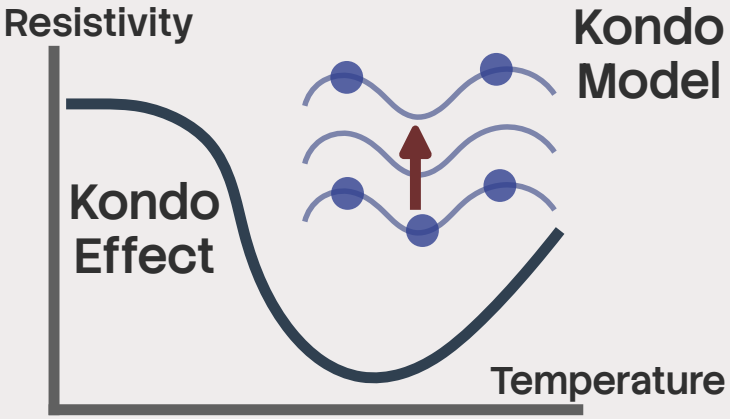
Chap 8. Conclusions

Introduction To Fermionic Criticality and Quantum Materials

What Do Condensed Matter Theorists Do?

Explain Experiments

- Build and solve theoretical models for experiments
- Compute measurables to explain observations
- Kondo effect, quantum Hall effect, superconductivity



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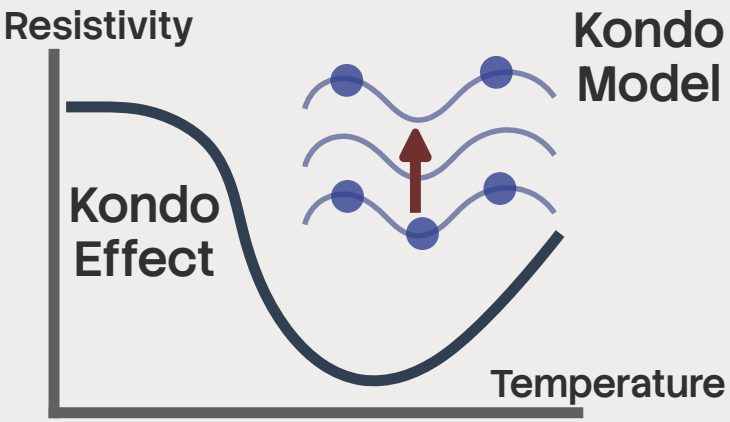
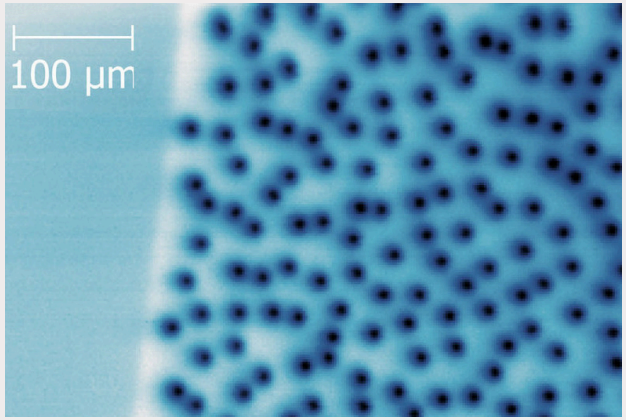


Predict Experiments

- Identify interesting theoretical models for new systems
- Predict novel phenomena by tweaking models and parameters
- Abrikosov vortex lattice, spin liquids



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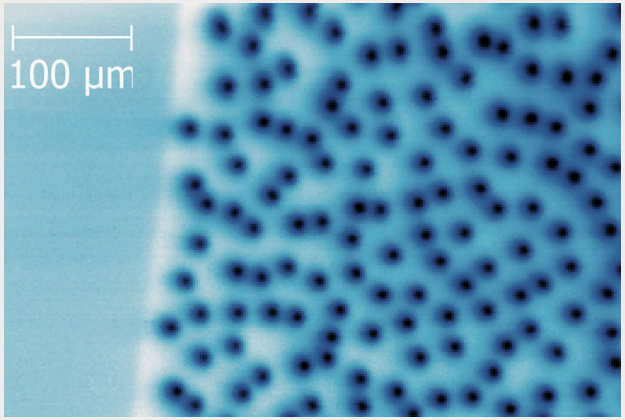

Classify experiments

- organize experimental signatures into universal principles
- come up with frameworks that explain a large class of phenomena
- Examples: Landau theory of phase transition

Quantum Materials

Class of materials that **cannot** be well-approximated by “classical” description

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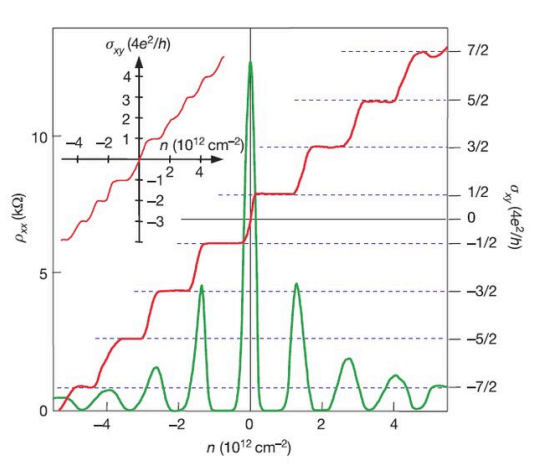

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(*topological insulators*)

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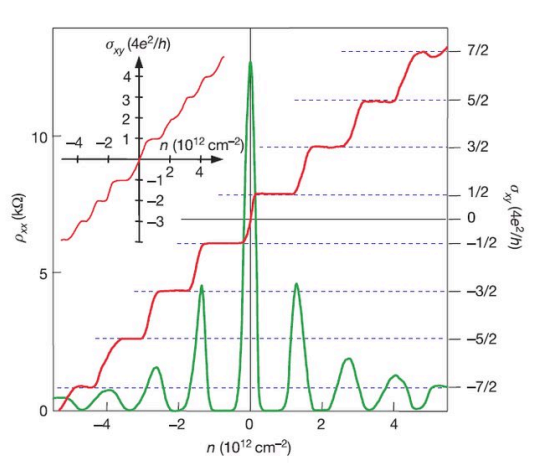

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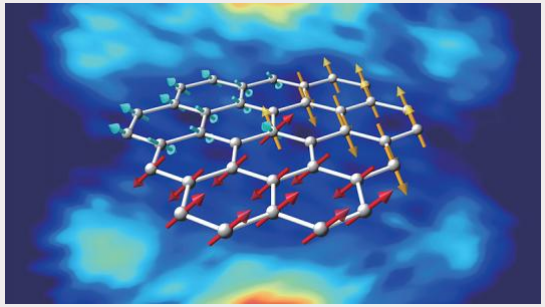

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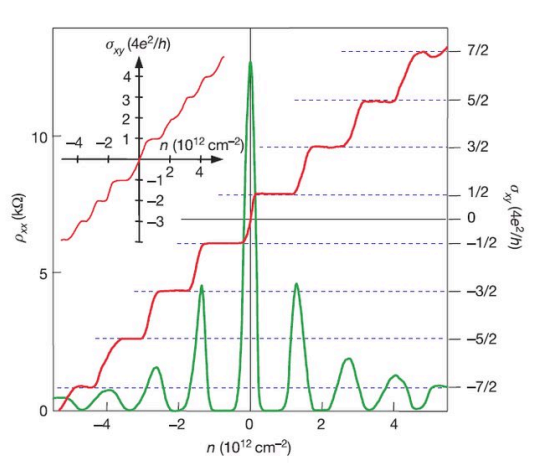

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collective excitations
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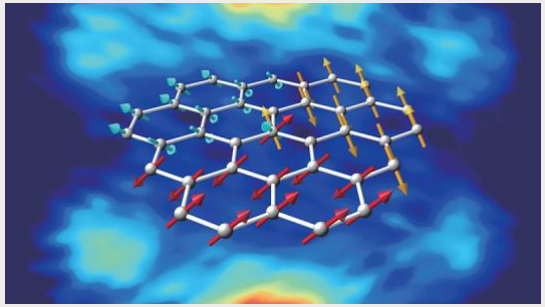

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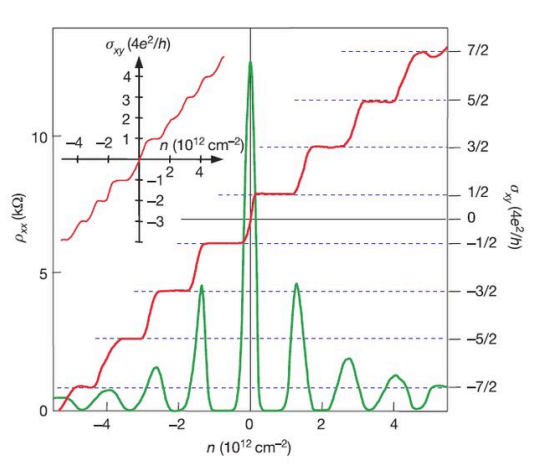

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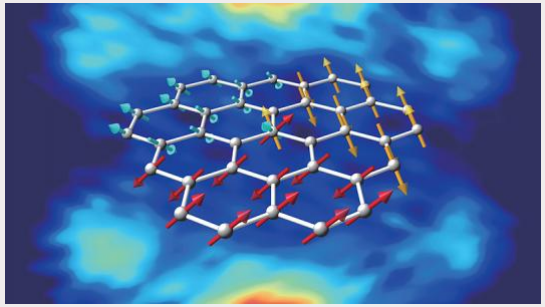

Needs dramatically **new ideas** and methods!

- **non-local/collective** order parameters for transition
- ability to work with a large number of **interacting particles**
- ability to work with **emergent** degrees of freedom

Strongly Correlated Phenomena

Materials with strong **local** electronic correlation, **low dimensionality**, multiple electronic **orbitals**

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unpaired *f* or *d* electrons

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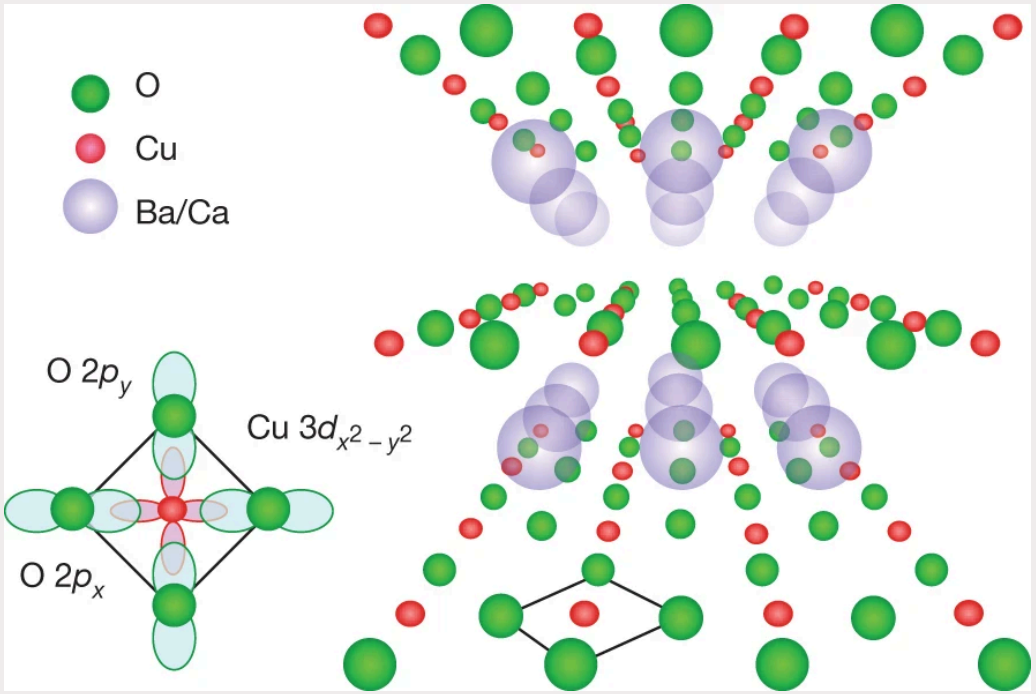

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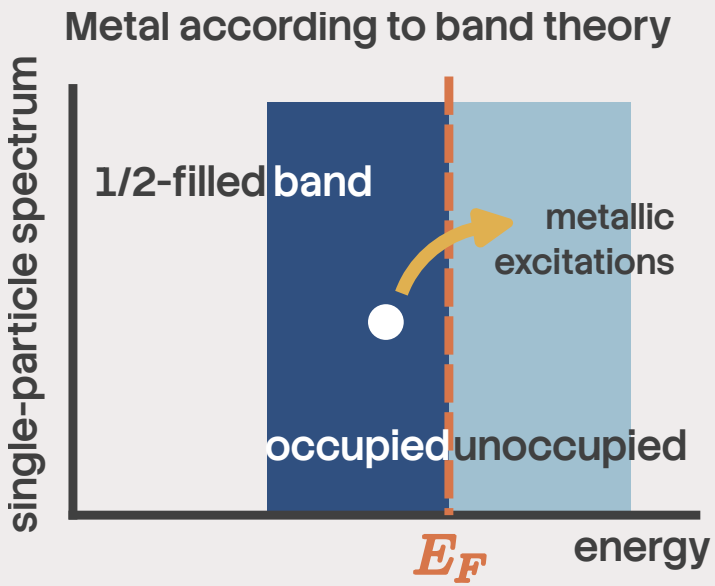

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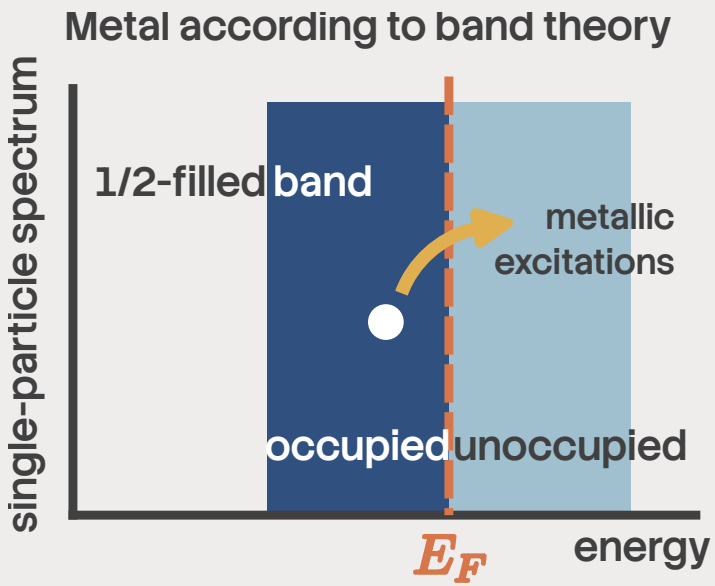

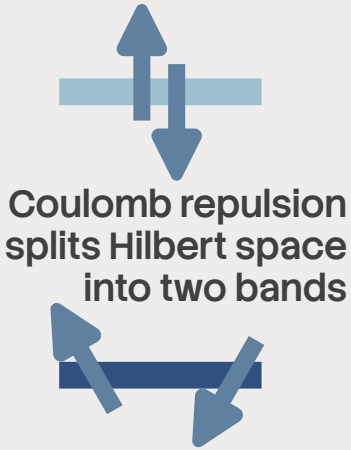

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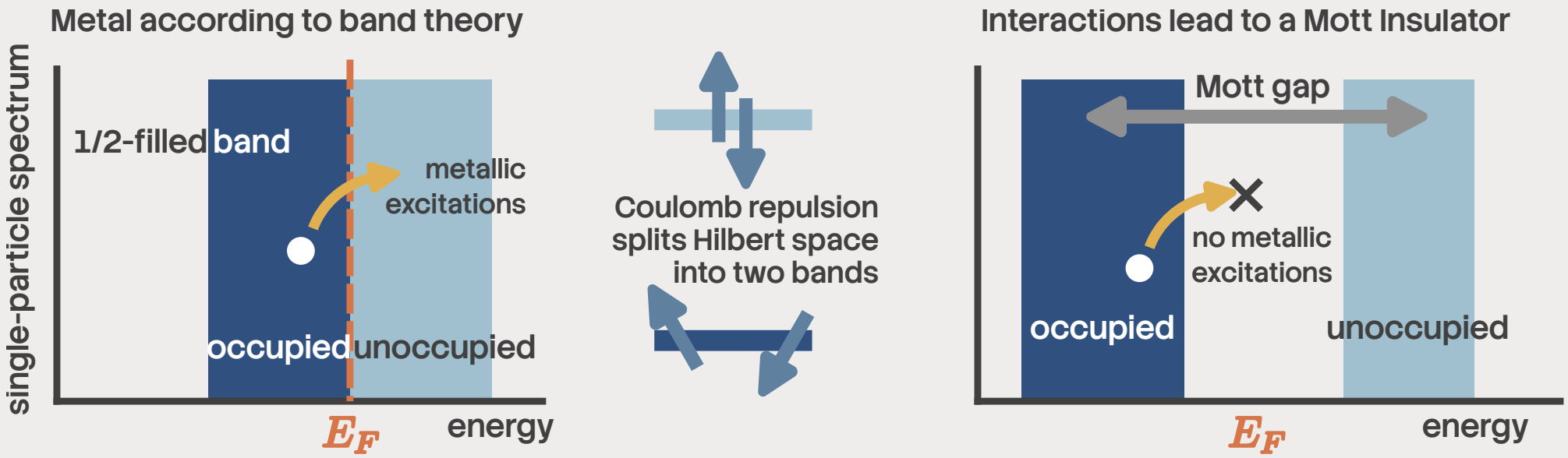

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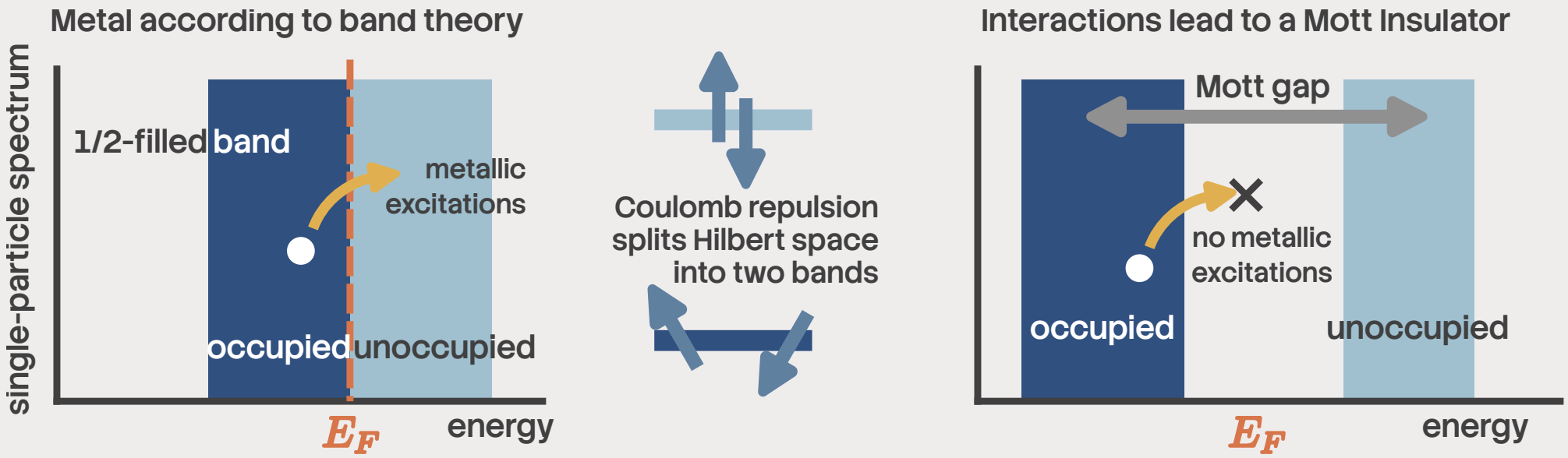

- Not accessible from 1-particle theory
- result of **manybody effects**

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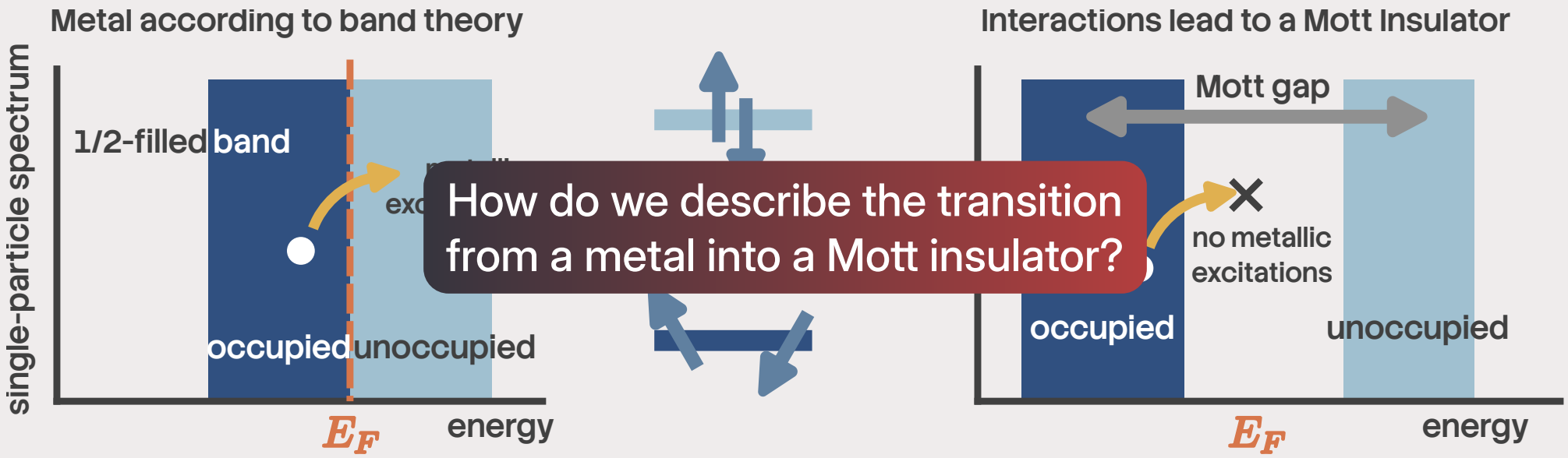

How do we describe the transition from a metal into a Mott insulator?

- Not accessible from 1-particle theory
- result of **manybody effects**

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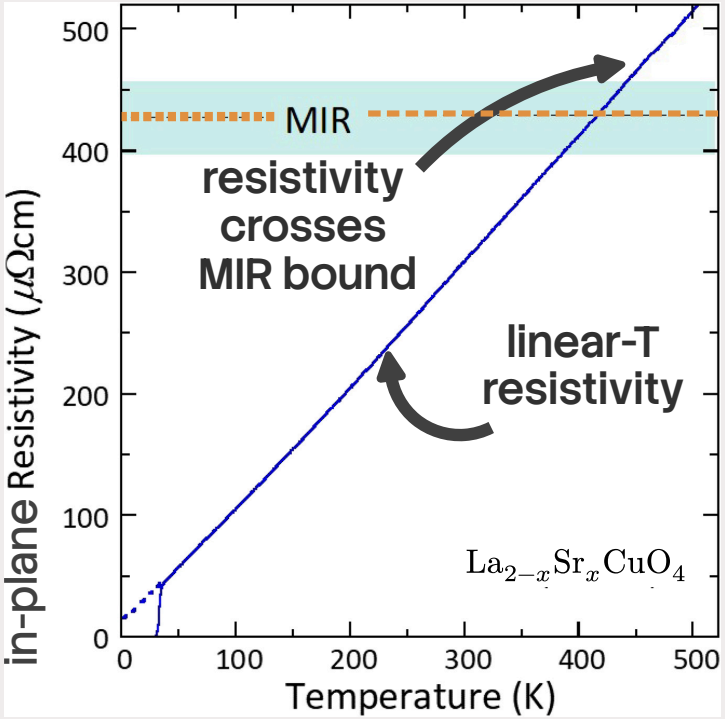
Phenomena: Strange Metallicity



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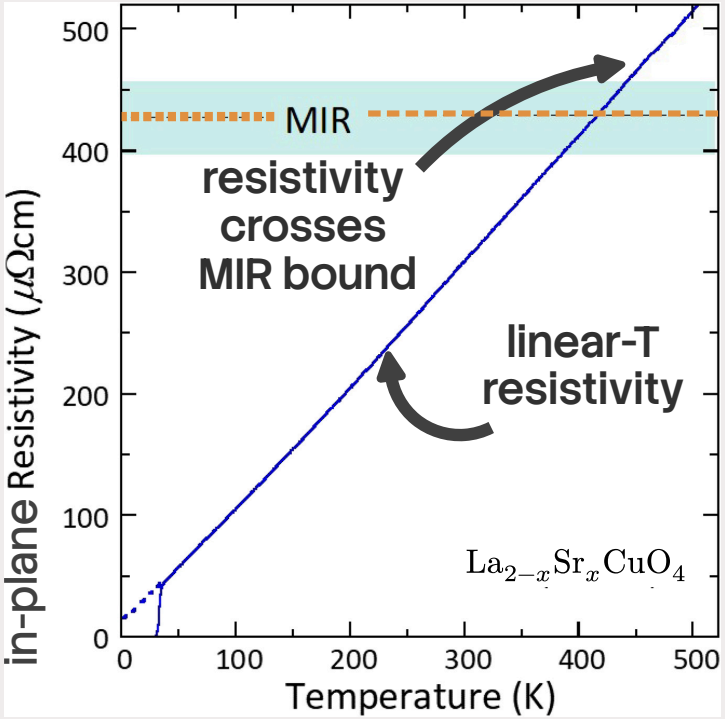
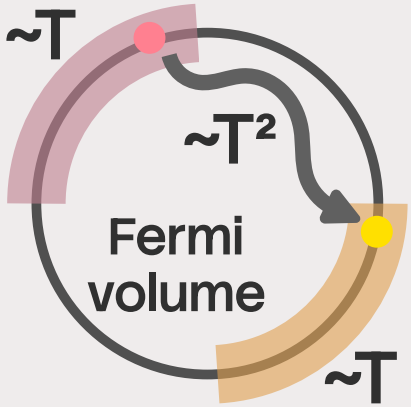


Conventional metallic theory predicts $R \propto T^2$



Source of $R \propto T^2$?

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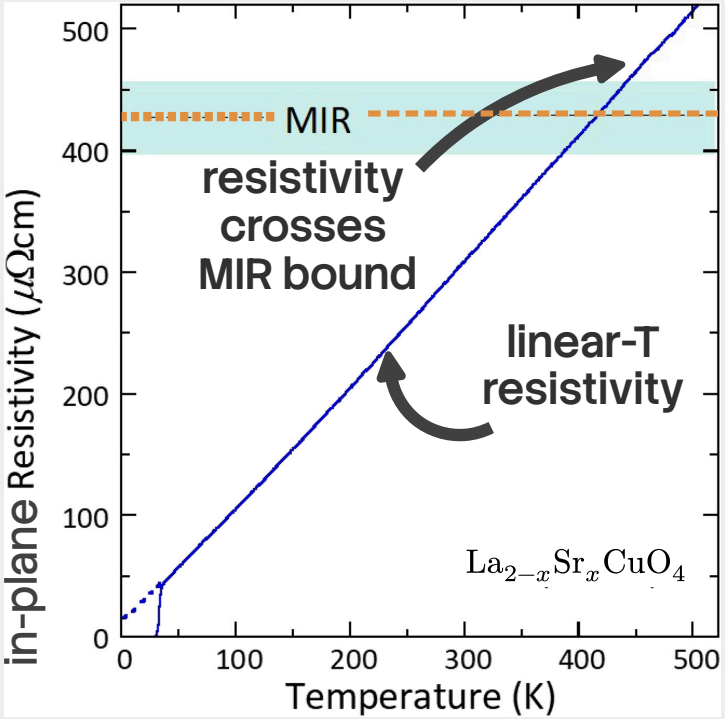

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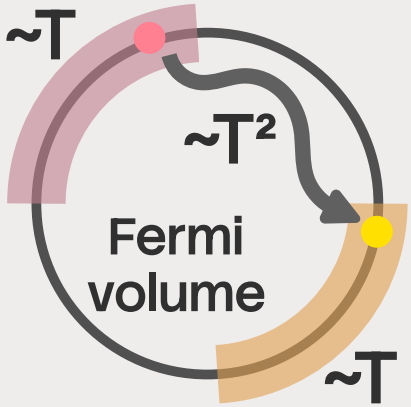
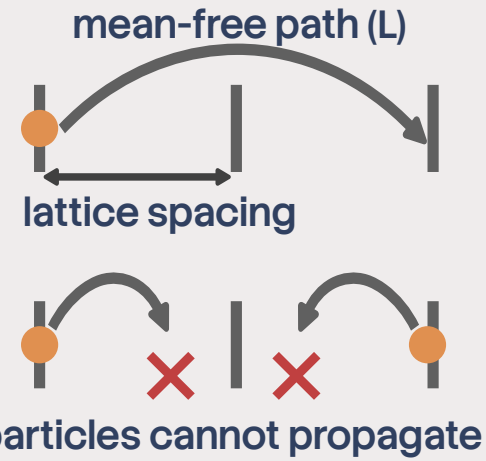


Source of $R \propto T^2$?

Single particles must propagate finite number of lattice spacing



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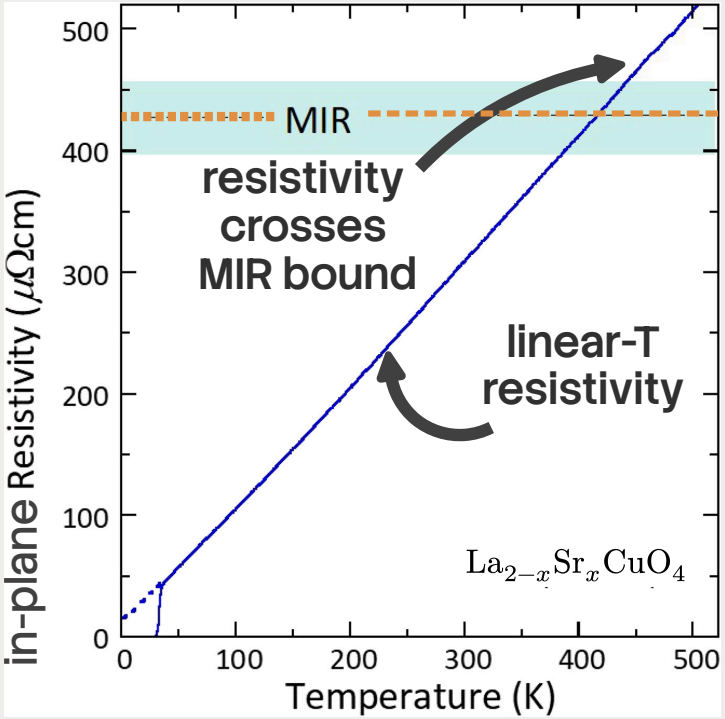

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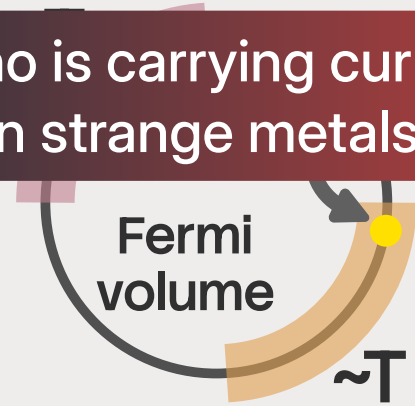
Who is carrying current in strange metals?



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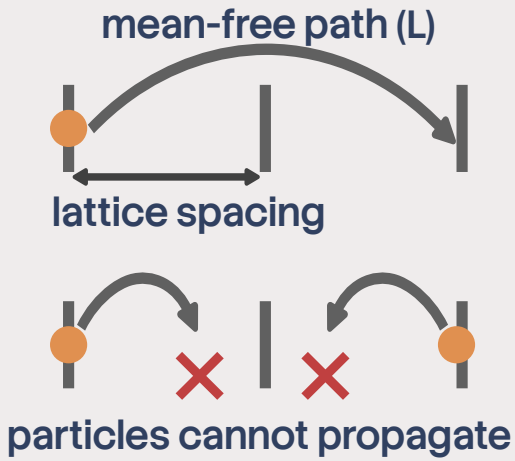
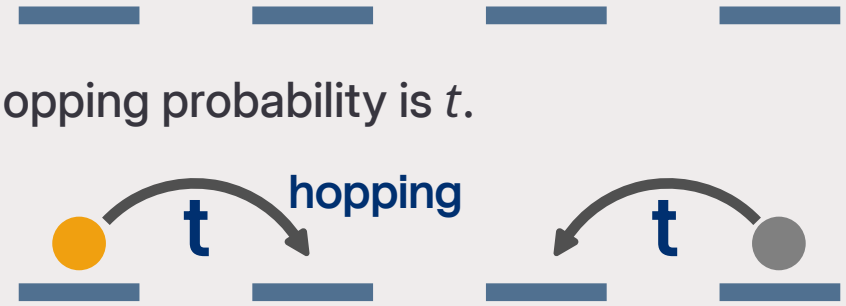


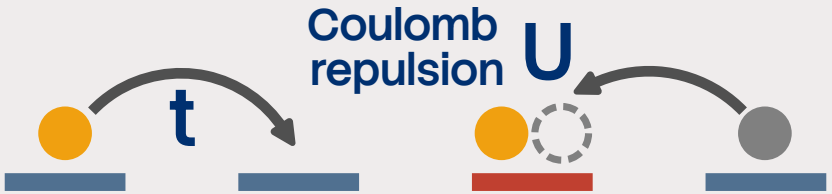
Theoretical Models of Correlated Materials

Hubbard Model

- Correlated electrons hopping on a lattice



- Hopping probability is t.
- Energy cost U if two occupy same site.



Hamiltonian: $-t\sum_{i,j,\sigma}(c^{\dagger}_{i,\sigma}c_{j,\sigma} + \text{h.c.}) + \sum_i U n_{i,\uparrow}n_{i,\downarrow}$

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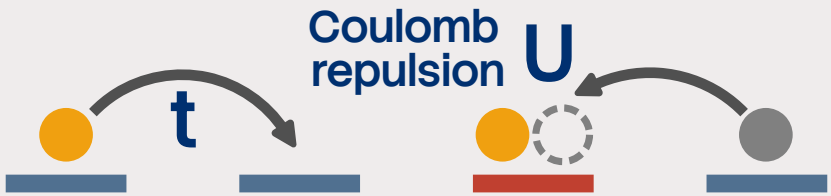


Quantum Impurity Models as Auxiliary Models

- Broad idea behind Auxiliary models
- What are quantum impurity models
- Example of mapping (mean-field) to previously mentioned models

Quantum Impurity Models as Auxiliary Models

More details

- discuss Kondo screening
- discuss Kondo breakdown
- argue how LFL and LM can be thought of as metallic vs insulating behaviour

Questions

- Questions about solving models
- Questions about developing methods

Minimal Model For Kondo Breakdown

A Route to Mott Localisation in Infinite Dimensions

New J. Phys. 25 113011 (2023)
AM, N S Vidhyadhiraja, A Taraphder, S Lal

Capturing Spatial Fluctuations and Lattice Geometry

Impurity-Lattice Mapping Beyond Mean-Field

New J. Phys. 28 013503 (2026)
AM, S R. Hassan, A Mukherjee, N S. Vidhyadhiraja, A Taraphder, S Lal

Kondo Breakdown On The Square Lattice

Revisiting the Mott Transition in Two Dimensions

New J. Phys. 28 013503 (2026)
AM, S R. Hassan, A Mukherjee, N S. Vidhyadhiraja, A Taraphder, S Lal

Competing tendencies In a Multi-Orbital System

Kondo screening vs. Mott localisation in a Heavy-Fermion model

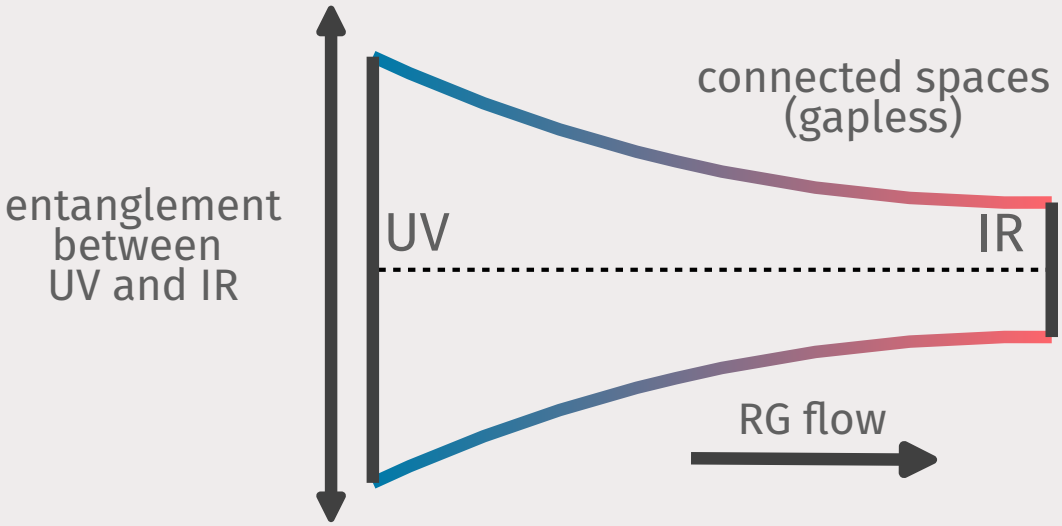
(in progress)

AM, S Lal

Holography of Entanglement in 2D Free Fermions

Emergent Geometry and Fermionic Criticality

J. Phys. A: Math. Theor. 57 275401 (2024)
AM, S Patra, S Lal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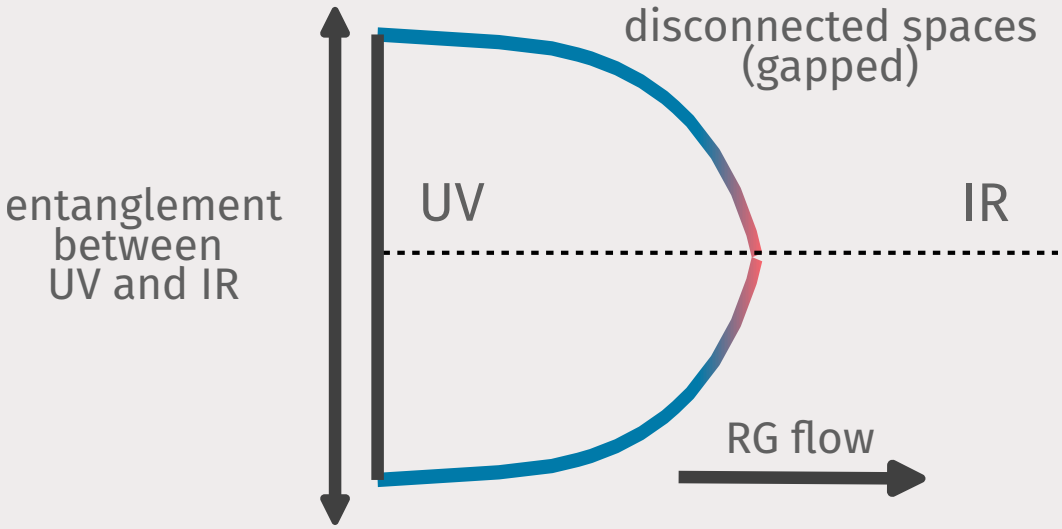
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Closing Remarks

Summary of Primary Results

Main Conclusions and Takeaways

Future Work